The Plant Journal (1993) **3**(6), 883–886

TECHNICAL ADVANCE

Localization of vicilin genes via polymerase chain reaction on microisolated field bean chromosomes

Jiri Macas†, Winfriede Weschke, Helmut Bäumlein, Uta Pich, Andreas Houben, Ulrich Wobus and Ingo Schubert*

Institute of Plant Genetics and Crop Plant Research, O-4325 Gatersleben, Germany

Summary

A new technique is reported for the physical mapping of low copy DNA sequences on plant chromosomes. Individual chromosomes were microisolated and their DNA used as the target for the polymerase chain reaction in order to identify the chromosome carrying a specific gene sequence. The use of defined translocation chromosomes further refined the resolution of the method to a subchromosomal level. To demonstrate the applicability of the procedure genes have been localized coding for vicilin seed storage proteins on the field bean *Vicia faba* L. in a region which includes the centromere and the proximal parts of the short and the long arms of chromosome II.

Introduction

Mapping of gene and marker sequences is a major goal of genetic research and plant breeding. For species in which segregating populations and extensive mapping data are already available, this can be achieved by genetic analysis. However, these prerequisites are not given for the majority of crop plants. In addition, mapping by genetic analysis can only proceed when the gene under consideration is polymorphic. Moreover, genetically detemined gene distances in some cases do not reflect the actual physical distances.

An alternative approach is physical sequence mapping by hybridization of labelled probes either to chromosomes *in situ* or to the DNA of chromosomes isolated in bulk. Although single copy sequences have been localized *in situ* on plant chromosomes (e.g. Ambros *et al.*, 1986; Huang *et al.*, 1988; Lehfer *et al.*, 1991), *in situ* hybridization is not yet a reliable technique for mapping short (~1 kb) single copy sequences.

Received 30 September 1992; revised 4 January 1993; accepted 18 January 1993.

*For correspondence (+49 39 482 280).

†Permanent address: Institute of Plant Molecular Biology, Branisovska 31, 37005 Ceske Budejovice, Czech Republic.

To overcome these problems we have developed a new approach. Individual chromosomes were isolated by micromanipulation from squash preparations of *Vicia faba* root tip meristems. To identify the chromosomes carrying a specific gene, the chromosomal DNA was used as the target for the polymerase chain reaction (PCR). By using defined translocation chromosomes from reconstructed karyotypes (Schubert and Rieger, 1991) the resolution of the technique was further improved to a subchromosomal level.

We have used this technique to localize the genes coding for a 50 kDa subunit of the 7S seed-storage protein, vicilin. The PCR primer sequences were derived from the genomic clone *Vfvic1* of vicilin (Weschke *et al.*, 1988). *Vfvic1* is a member of the small vicilin gene family composed of about five members according to Southern hybridization data (Wobus, unpublished data) and polypeptide polymorphism (Müntz *et al.*, 1986).

The detection of an amplified fragment of the expected size was reproducibly correlated in the presence of the proximal part of the short and long arms of chromosome II including the centromere.

This procedure is suitable especially for plant species where conventional methods for the localization of single or low copy sequences are not applicable, and where the gene under study does not exhibit polymorphism in the sense of allelism.

Results and discussion

Chromosomal location of vicilin genes

In model reactions we first established conditions which allowed the specific amplification of a vicilin gene fragment. For that purpose we used a primer pair spanning the *Vfvic1* sequence from position +42 at the G of the ATG translation initiation codon to position +882 (including the first two introns), and as targets we used either the *Vfvic1* clone (0.5 pg), total genomic DNA (125 pg), or two to ten microisolated prophase nuclei. In all cases a band of the expected size of 841 bp was visible on the gel after 40 cycles of amplification.

We then selected eight different types of individually distinguishable translocation chromosomes (Figure 1) from plants with reconstructed karyotypes to localize the sequence within defined intrachromosomal regions.

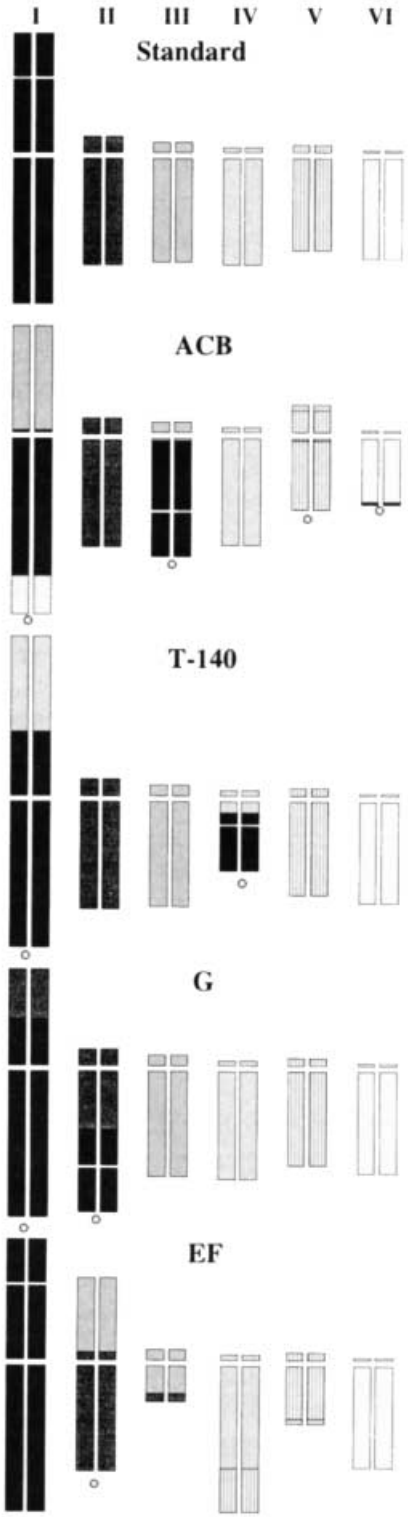


Figure 1. Scheme of the standard (wild-type) and of reconstructed karyotypes of *V. faba*.
Chromosomes as targets for PCR are marked by ○. Karyotype ACB: chromosome I composed of parts of wild-type chromosomes I, III and VI; chromosome III composed of parts of chromosomes I and III; the pericentrically inverted chromosome V; the small chromosome VI with a terminal piece of the long arm of wild-type chromosome I. Karyotype T140: translocation chromosomes I and IV. Karyotype G: translocation chromosomes I and II. Karyotype EF: translocation chromosomes II and III (chromosome II used separately for PCR; chromosome III together with the other chromosomes of karyotype EF used as a pooled control).

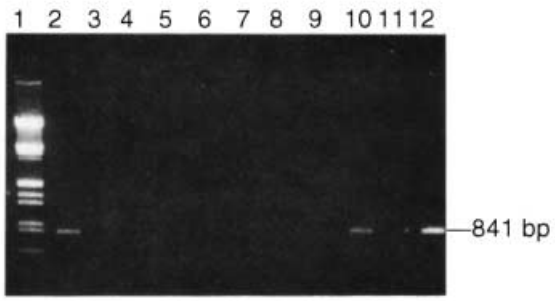


Figure 2. Agarose gel electrophoresis of PCR products obtained with vicilin specific primers.
Lane 1, size markers (*Eco*RI/*Hin*dIII digested lambda DNA); lane 2, 125 pg total genomic DNA of *V. faba*; lanes 3–6 chromosomes I, III, V, and VI of karyotype ACB; lanes 7 and 8, chromosomes I and IV of karyotype T140; lanes 9 and 10, chromosomes I and II of karyotype G; lane 11, no DNA; lane 12, 0.5 pg or *Vfvic1* DNA.

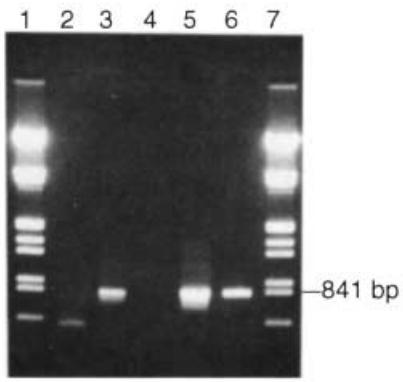


Figure 3. Agarose gel electrophoresis of PCR products obtained with vicilin specific primers.
Lanes 1 and 7, size markers (*Eco*RI/*Hin*dIII digested lambda DNA); lane 2, all chromosomes (20 each) of karyotype EF except chromosome II; Lane 3, chromosome II of karyotype EF; lane 4, no DNA; lane 5, 125 pg total genomic DNA of *V. faba*; lane 6, 0.5 pg of *Vfvic1* DNA. The approximate 500 bp band in lane 2 is an artifact which occasionally appeared after 40 cycles of PCR.

When DNA of chromosomes I, III, V, VI of karyotype ACB, I, IV of karyotype T140 or I, II of karyotype G (Figure 1) were separately used as targets for PCR, only chromosome II of karyotype G yielded a band of 841 bp (Figure 2). Since chromosomes I and III of karyotype ACB and I and IV of karyotype T140 did not yield a band, either the short arm or the proximal half of the long arm of wild-type chromosome II, as original parts of chromosome GII, should bear the site of the *Vfvic1* gene.

To confirm and refine this result a second series of experiments was run with karyotype EF in which about half of the short arm of chromosome II is substituted by most of the long arm of chromosome III. PCR was done separately with DNA of 20 chromosome IIs and of the remaining five chromosome types (20 of each) pooled together. Again only DNA of chromosome II yielded a 841 bp band (Figure 3), which was identified as vicilin sequence by Southern

hybridization with a genomic vicilin gene probe (data not shown).

The vicilin sequence between the primers contains two *Bgl*II recognition sites cleaving out a 109 bp fragment between positions 628 and 736 of *Vfvic1*.

This fragment was cloned and sequenced and proved to be identical to the corresponding *Vfvic1* sequence (data not shown).

The PCR data obtained with DNA of different chromosomes indicate that all vicilin genes which can be detected by the chosen primer pair under the described conditions are located within a region including the proximal halves of the short and the long arms of chromosome II.

Comparison with other gene mapping methods

As stated above, DNA of two 4C nuclei resulted, after amplification, in a detectable band corresponding to the vicilin sequence.

Given the presence of approximately five copies of the amplified sequence per haploid (1C) genome, about 40 respective molecules are detectable by this method. Thus, for detection of even single copy genes 20 metaphase chromosomes (40 copies) are sufficient.

This is less than the 50–200 flow-sorted human chromosomes used by Cotter *et al*. (1989) to detect single copy genes by means of PCR.

When compared with *in situ* hybridization, we conclude that PCR on isolated chromosomes is presently more sensitive for detection of single copy sequences, since *in situ* hybridization still needs targets of several kilobases to yield a reliable signal. The precision of localization is comparable with that obtained by *in situ* hybridization when suitable translocation chromosomes can be used. Our technique is less laborious and time-consuming than genetic linkage analysis, especially in plant species such as *V. faba* where genetic maps are not well developed. A further advantage is the ability to localize chromosomal position of genes without allelic polymorphism.

Experimental procedures

Isolation of defined Vicia faba *chromosomes*

Metaphase chromosomes of *Vicia faba* ($n = 6$) were isolated from the following reconstructed karyotypes (Figure 1; for detailed description see Schubert and Rieger, 1991): ACB (chromosomes, I, III, V and VI), T140 (I and IV), G (I and II) and EF (II). These chromosomes are not only easily distinguishable by their length and/or morphology (Figure 1) but additionally they allow the localization of sequences within defined intrachromosomal regions (the translocated segments) of wild-type chromosomes. Chromosomes were isolated from squash preparations of colchicine-arrested root meristems as described by Schubert *et al*. (1993). Cover-slips with squashes were stored in glycerol at 4°C before use. Glycerol was removed by several washes in distilled water and chromosomes were stained for 10 sec in 0.01% methylene blue.

For microscopically controlled chromosome isolation a system employing an oil chamber and a mechanically driven sliding micromanipulator (Carl Zeiss, Jena) was used. Defined chromosomes were picked up by a glass needle and transferred into a collection drop hanging underneath a siliconized cover-slip. The collection drop contained 0.3–0.5 µl of proteinase K (Boehringer, 0.5 mg ml^{-1}) in 10 mM Tris-HCl pH 7.5, 10 mM NaCl and 0.1% SDS (Lüdecke *et al*., 1989). Twenty chromosomes were pooled and the drop transferred together with a small volume of oil into a PCR reaction tube containing 50 µl of PCR mix.

Conditions of the polymerase chain reaction

Primers were designed according to the genomic vicilin sequence (Weschke *et al*., 1988). The forward primer was a 30-mer (5′-GGCAGCTACCACATTGAAAGATTCATTTCC-3′), the reverse primer a 32-mer (5′-CCAGATAAGATGGACGGTTGGTTTTGATTACC-3′).

The final concentration of the individual compounds in the PCR mix was as follows: 0.2 mM of each dNTP, 0.2 µM primers, 2.0 mM $MgCl_2$, 50 mM KCl, 10 mM Tris-HCl pH 9.0 at 25°C, 0.1% Triton X-100. Mixtures were overlaid by mineral oil and after initial denaturation (94°C 10 min) the samples were cooled to 85°C and 2.5 units of Taq polymerase (Promega) were added. Then 40 cycles of amplification (primer annealing: 55°C 0.8 min; primer extension: 72°C 1 min; denaturation: 94°C 0.8 min) ending with a final primer extension for 5 min were performed in an Autogene II (Grant) thermal cycler. Genomic DNA (125 pg), DNA of the genomic clone *Vfvic1* (0.5 pg, 27 cycles only) or one to ten microisolated nuclei were used as a control. Amplification products were electrophoresed on 1% agarose gels.

Acknowledgements

We are grateful to Dr B. Schlesier and R.Gillandt for synthesizing primers, to B. Hildebrandt for preparation of slides, and to Prof. R. Rieger and Dr G. Ramsay for critical reading of the manuscript. This work was supported by the Deutsche Forschungsgemeinschaft.

References

Ambros, P., Matzke, M.A. and Matzke, A.J.M. (1986) Detection of a 17 kb unique sequence (T-DNA) in plant chromosomes by *in situ* hybridization. *Chromosoma*, **94**, 11–18.

Cotter, F., Nasipuri, S., Lam, G. and Young, B.D. (1989) Gene mapping by enzymatic amplification from flow-sorted chromosomes. *Genomics*, **5**, 470–474.

Huang, P.-L., Hahlbrock, K. and Somssich, I.E. (1988) Detection of a single-copy gene on plant chromosomes by *in situ* hybridization. *Mol. Gen. Genet*. **211**, 143–147.

Lehfer, H., Wanner, G. and Herrmann, R.G. (1991) Physical mapping of DNA sequences on plant chromosomes by light microscopy and high resolution scanning electron microscopy. In *Plant Molecular Biology*, Volume 2 (Herrmann, R.G. and Larkins, G., eds). New York: Plenum Press, pp. 277–284.

Lüdecke, H.-J., Senger, G., Claussen, U. and Horsthemke, B. (1989) Cloning defined regions of the human genome by microdissection of banded chromosomes and enzymatic amplification. *Nature*, **338**, 348–350.

Müntz, K., Horstmann, C. and Schlesier, B. (1986) Seed proteins and their genetics in *Vicia faba* L. *Biologisches Zentralblatt*, **105**, 107–120.

Schubert, I. and Rieger, R. (1991) Catalogue of chromosomal and morphological mutants of *Faba* bean in the Gatersleben Collection, 1991. *FABIS Newslett.* **28/29**, 14–22.

Schubert, I., Dolezel, J., Houben, A., Scherthan, H. and Wanner, G. (1993) Refined examination of plant metaphase chromosome structure at different levels made feasible by new isolation methods. *Chromosoma*, **102**, 96–101.

Weschke, W., Bassüner, R., Hai, N.V., Czihal, A., Bäumlein, H. and Wobus, U. (1988) The structure of a *Vicia faba* vicilin gene. *Biochem. Physiol. Pflanzen* **183**, 233–242.